NAME OF CANDIDATE: .....
STUDENT NUMBER: .....
COURSE (3871 or 5960): .....
TUTORIAL GROUP (day & time): .....

THE UNIVERSITY OF NEW SOUTH WALES

SCHOOL OF MATHEMATICS AND STATISTICS

Mid-Session Test, T3 2025

MATH3871/5960

Bayesian Inference and Computation

(1) TIME ALLOWED – 1.5 hours + 10 mins reading time

(2) TOTAL NUMBER OF QUESTIONS – 4

(3) ANSWER ALL QUESTIONS

(4) THE QUESTIONS ARE OF EQUAL VALUE

(5) WRITE YOUR ANSWERS IN THE SPACE PROVIDED

(6) THIS PAPER MAY **NOT** BE RETAINED BY THE CANDIDATE

(7) ONLY CALCULATORS WITH AN AFFIXED "UNSW APPROVED" STICKER MAY BE USED.

All answers must be written in ink. Except where they are expressly required pencils may only be used for drawing, sketching or graphical work.

1. **[10 marks]** Let x be the number of successes in n independent trials with probability p of success in each trial. Suppose that both the probability of success p is unknown and the number of trials n is unknown (i.e. the number of trials that can be conducted depends on some random process). The likelihood function is $L(x|n,p) = \binom{n}{x}p^x(1-p)^{n-x}$. Let n and p have independent prior distributions, i.e. $\pi(n,p) = \pi(n)\pi(p)$. Suppose both $\pi(n)$ and $\pi(p)$ are uniform priors (i.e. $\pi(n) \propto 1$ and $\pi(p) \propto 1$).

Hint: Recall that for positive integers n, x,

$$\binom{n}{x} = \frac{n!}{x!(n-x)!} \qquad \textit{and} \qquad \Gamma(n) = (n-1)!$$

a) **[1 marks]** Are $\pi(n)$ and $\pi(p)$ proper or improper priors? Explain why.

b) **[3 marks]** Give the joint posterior distribution $\pi(n,p|x)$ and the marginal posterior distribution $\pi(n|x)$. You may provide your answers in unnormalised form.

c) **[2 marks]** Based on your answer in (b), explain whether the chosen prior $\pi(n)$ is acceptable to use or not.

d) **[2 marks]** Suppose that

$$\hat{\pi}(\theta|x) = \frac{1}{N}\sum_{i=1}^{N}\delta_{\theta^{(i)}}(\theta)$$

is an approximation of the posterior distribution, where $\theta^{(1)}, \ldots, \theta^{(N)} \sim \pi(\theta|x)$, and $\delta_b(A) = 1$ if $b \in A$ and 0 otherwise. Show that an approximation to an expectation with respect to the posterior is given by

$$E_\pi[g(\theta)] \approx \frac{1}{N}\sum_{i=1}^{N} g(\theta^{(i)}).$$

e) **[2 marks]** Suppose that $\theta^{(1)}, \ldots, \theta^{(N)} \sim \pi(\theta|x)$, and that $\bar{g} = \frac{1}{N}\sum_{i=1}^{N} g(\theta^{(i)})$ and $s_g^2 = \frac{1}{N}\sum_{i=1}^{N}(g(\theta^{(i)}) - \bar{g})^2$. How many samples, $g(\theta^{(i)})$, are required to ensure that the Monte Carlo estimate, $\bar{g}$, is within 0.01 of its true value with probability $1-\alpha$, for $\alpha \in (0,1)$?

Please see over ...

2. [**10 marks**]

a) [**3 marks**] Determine the form of a conjugate prior for the exponential family of models, given by

$$f(x|\theta) = h(x)g(\theta)\exp\{t(x)c(\theta)\},$$

for functions h, g, t, and c such that $\int f(x|\theta) = 1$.

b) [**2 marks**] Hence, using the result from part a), *derive* the form of the conjugate prior when $X \sim \text{Binomial}(n, \theta)$.

c) [**1 marks**] When is it appropriate and when is it not appropriate to use conjugate priors?

d) [**1 marks**] Do conjugate priors exist for a model $h(x|\theta)$ in *any* family? Explain why. If so, are there any conditions?

e) [**2 marks**] Suppose that X is a random variable with distribution function $F(x)$. By calculating the cdf, show that the distribution of $U = F(X)$ is U(0,1). Hence, deduce a method of generating samples from $F(x)$ given draws $u^{(1)}, \ldots, u^{(N)} \sim U(0, 1)$.

f) [**1 marks**] Give one benefit and one weakness of inversion sampling.

Please see over ...

3. [10 marks]

a) **[3 marks]** By using a suitable transformation that maps the integration limits to $[0, 1]$, write the integral $\int_a^\infty g(x)dx$, for $0 \leq a \leq \infty$, in the form $\int_0^1 g'(u)du$. Hence write down a Monte Carlo estimate of this integral using samples $u^{(1)}, \ldots, u^{(N)} \sim U(0, 1)$.

b) **[2 marks]** The rejection sampling algorithm can be written in two ways (given below). By considering the overall acceptance probability of each algorithm, explain why these two versions are equivalent.

Rejection sampling (version 1)

i) Generate x^* from $g(x)$
ii) Generate y^* from $U(0, Kg(x^*))$
iii) Accept x^* if $y^* \leq f(x^*)$

Rejection sampling (version 2)

i) Generate x^* from $g(x)$
ii) Accept x^* with probability $\frac{f(x^*)}{Kg(x^*)}$

where in each case, $f(x) \leq Kg(x)$, $\forall x$.

c) **[3 marks]** Let $X \sim g(x)$ such that $f(x) \leq Kg(x)$, $\forall x$, for some K. By evaluating $\Pr(X \leq x | X \text{ is accepted})$, show that samples accepted under rejection sampling follow the distribution $f(x)$. Assume that $\int f(y)dy = c < \infty$.

d) **[2 marks]** Hence, or otherwise, calculate the overall acceptance rate of the algorithm. How does the overall acceptance rate inform the optimal choice of K?

Please see over ...

4. [10 marks]

a) **[3 marks]** Show that the expected posterior loss,

$$E_\pi[L(\theta, d)] = \int L(\theta, d)\pi(\theta|x)d\theta$$

is minimised at the posterior mean $d = E(\theta|x)$ when $L(\theta, d) = (\theta - d)^2$ is the quadratic loss function.

b) **[2 marks]** In a particular Bayesian analysis, where the loss function quantifies the \$ value in costs of some industrial process, the posterior variance is estimated as $\hat{V} = 2.8$ million. Under quadratic loss, if a company is willing to tolerate expected costs of \$C or less, what range of point estimates, d, can they use?

c) **[1 marks]** Can the company achieve their goal if \$C = \$2.3 million?

d) **[3 marks]** Suppose that $x_1, \ldots, x_n$ follow the Pareto distribution, Pareto(a, b), with known a, which has density function

$$f(x|b) = ba^b x^{-b-1} \qquad \text{for } x > a.$$

Find the asymptotic posterior distribution for b.

e) **[1 marks]** How would the asymptotic posterior in part d) change for different prior distributions for b?

Please see over ...

Some useful definitions and results

- If $X \sim \text{Beta}(\alpha, \beta)$, $\alpha > 0, \beta > 0$, $x \in (0,1)$, $p(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)}$. $E(X) = \alpha/(\alpha+\beta)$, $Var(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$.
- If $X \sim \text{Gamma}(\alpha, \beta)$, $\alpha, \beta > 0, x \geq 0$, $p(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$. $E(x) = \alpha/\beta$, $Var(X) = \alpha/\beta^2$.
- For positive integers α, $\Gamma(\alpha) = (\alpha-1)!$
- If $X \sim \text{Binomial}(n, p)$, $p \in [0,1]$, $P(X = k) = \left(\begin{array}{c} n \\ k \end{array} \right) p^k (1-p)^{n-k}$, for $k = 0, \ldots, n$. $E(X) = np$, $Var(X) = np(1-p)$.
- If $X \sim \text{Poisson}(\lambda), \lambda > 0$, $P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$ for $k = 0, \ldots,$. $E(X) = \lambda$, $Var(X) = \lambda$.
- If $X \sim N(\theta, \sigma^2)$, $p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp(-\frac{1}{2\sigma^2}(x-\theta)^2)$, $E(X) = \theta, Var(X) = \sigma^2$.
- If $X \sim \text{Uniform}(a, b), b > a$, $p(x) = 1/(b-a)$, $E(X) = \frac{1}{2}(a+b)$, $Var(X) = \frac{1}{12}(b-a)^2$.
- If $X \sim Exp(\lambda)$ with $\lambda > 0$, $p(x) = \lambda \exp(-\lambda x), \quad x \geq 0$, $E(X) = \frac{1}{\lambda}$ and $Var(X) = \frac{1}{\lambda^2}$.

Some standard conjugate prior-likelihood pairs.

Likelihood	Prior	Posterior
$x \sim Binomial(n, \theta)$	$Beta(p, q)$	$Beta(p + x, q + n - x)$
$x_1, \ldots, x_n \sim Poisson(\theta)$	$Gamma(p, q)$	$Gamma(p + \sum_{i=1}^n x_i, q + n)$
$x_1, \ldots, x_n \sim N(\theta, \tau^{-1})$, ($\tau$ known)	$N(b, c^{-1})$	$N(\frac{cb+n\tau\bar{x}}{c+n\tau}, \frac{1}{c+n\tau})$
$x_1, \ldots, x_n \sim Gamma(k, \theta)$ (k known)	$Gamma(p, q)$,	$Ga(p + nk, q + \sum_{i=1}^n x_i)$
$x_1, \ldots, x_n \sim Geometric(\theta)$	$Beta(p, q)$	$Beta(p + n, q + \sum_{i=1}^n x_i - n)$
$x \sim NegBinomial(r, \theta)$	$Beta(p, q)$	$Beta(p + r, q + x)$